Macroeconomics in the Sequence Space

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Sequence Space vs. State Space

Sequence Space vs. State Space

Motivation

- In heterogeneous-agent (HA) models, the economy's state is the entire cross-sectional distribution — an infinite-dimensional object.
- Sequence-space methods *sidestep* this challenge: they characterize local dynamics around the steady state without ever carrying the distribution as a state variable.
- The key insight: aggregate equilibrium responses to shocks can be encoded in Jacobian matrices of aggregate outcomes with respect to aggregate inputs.

Sequence Space vs. State Space

State Space

Definition 1

A state-space representation describes the economy recursively. At date t, a state variable x_t summarizes all payoff-relevant history, and the model evolves as

$$x_{t+1} = f(x_t, \varepsilon_{t+1}), \qquad y_t = g(x_t).$$

Given x_t, the future is independent of the past.

Definition 2

A sequence-space representation characterizes equilibrium as a mapping between entire time paths. We seek sequences $\{X_t\}_{t=0}^{\infty}$ such that equilibrium conditions hold at every date:

$$F(\{X_t\}_{t=0}^{\infty}, \{Z_t\}_{t=0}^{\infty}) = 0,$$

where $\{X_t\}$ collects endogenous variables and $\{Z_t\}$ collects exogenous shocks or policy paths.

Sequence Space vs. State Space

Comparison

Intuition

- State space asks: *what is the current state, and how does it transition?*
- Sequence space asks: *given a shock path, what equilibrium outcome path clears all markets?*

Takeaway

When the distribution is the intractable state variable, sequence space reframes a "huge state" problem as a structured Jacobian problem — amenable to fast, scalable computation.

SSJ in Krusell-Smith Model

Example: The Krusell-Smith Model

Setup

- The economy is populated by a mass 1 of heterogeneous households,
- Each household maximizes its expected lifetime utility:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right],$$

- Standard CRRA utility,
- There exist n_e idiosyncratic states w/ transition matrix $P(e, e')$ and stationary distribution π,
- Agents supply an exogenous number of hours n, and earn wage income $w_t e n$,
- Agents can only trade in capital k, which pays a rental rate r_t net of depreciation.

Example: The Krusell-Smith Model

Household Problem

The value function of an agent entering period w/ state (e, k_-) is given by:

$$V_t(e, k_-) = \max_{c,\, k} u(c) + \beta \sum_{e'} V_{t+1}(e', k)\, P(e, e')$$

$$\text{s.t.} \qquad c + k = (1 + r_t)k_- + w_t e n$$

$$k \geq 0$$

- Denote by $c_t^*(e, k_-)$ and $k_t^*(e, k_-)$ the policy functions that solve the Bellman equation.
- Also denote by $D_t(e, K_-)$ the measure of households in state e that own capital in a set K_- at the start of date t.

Example: The Krusell-Smith Model

Household Problem

The distribution D_t has law of motion

$$D_{t+1}(e', K) = D_t(e, k_t^{*-1}(e, K))P(e, e').$$

where $k_t^{*-1}(e, K)$ denotes the inverse of $k_t^*(e, K)$.

- We assume that prior to $t = 0$, the economy is in a steady state.
- We suppose that agents start in this stationary distribution at date 0, so that $D_0 = D_{ss}$.

Example: The Krusell-Smith Model

Aggregate Capital

- For any t, the policy $k_t^*(e, k_-)$ is a function of the future path $\{r_s, w_s\}_{s=t}^{\infty}$.
- Given $D_0 = D_{ss}$, the distribution $D_t(e, K)$ at any t is a function of the entire path $\{r_s, w_s\}_{s=0}^{\infty}$.
- The aggregate household capital holdings are characterized by a capital function $\mathcal{K}_t(\{r_s, w_s\}_{s=0}^{\infty})$.

$$\mathcal{K}_t\left(\{r_s, w_s\}_{s=0}^{\infty}\right) = \sum_e \int_{k_-} k_t^*(e, k_-) D_t(e, dk_-)$$

Key Insight

The ability to reduce interactions between heterogeneous agents to functions such as $\mathcal{K}_t$, which map aggregate sequences into aggregate sequences, is key to the sequence-space Jacobian method.

Example: The Krusell-Smith Model

Production

- Firm adopts a Cobb-Douglas production function:

$$Y_t = Z_t K_t^{\alpha} N_t^{1-\alpha}$$

- The firm's first-order conditions:

$$r_t = \alpha Z_t K_{t-1}^{\alpha-1} N_t^{1-\alpha} - \delta$$
$$w_t = (1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}$$

Key Insight

The paths of prices $\{r_t, w_t\}_{t=0}^{\infty}$ are now functions of the exogenous paths $\{Z_t, N_t\}_{t=0}^{\infty}$ and the endogenous path $\{K_t\}_{t=0}^{\infty}$.

Example: The Krusell-Smith Model

Capital Market-Clearing Condition

Combining together, we can express the capital market-clearing condition at each point in time as a function H_t:

$$H_t(\mathrm{K}, \mathrm{Z}) \equiv \mathcal{K}_t \left(\left\{ \underbrace{\alpha Z_s \left(\frac{K_{s-1}}{\sum \pi(e) en} \right)^{\alpha-1} - \delta}_{r_s}, \underbrace{(1-\alpha) Z_s \left(\frac{K_{s-1}}{\sum \pi(e) en} \right)^{\alpha}}_{w_s} \right\}_{s=0}^{\infty} \right) - K_t = 0$$

where K is the vector of capital holdings $\{K_t\}_{t=0}^{\infty}$ and Z is the vector of exogenous shocks $\{Z_t\}_{t=0}^{\infty}$.

Equilibrium Capital Path

Given initial capital K_{-1} and the exogenous path for productivity Z , we have equilibrium path of capital $\{K_t\}_{t=0}^{\infty}$ by solving the nonlinear equation $H_t(\mathrm{K}, \mathrm{Z}) = 0$.

Example: The Krusell-Smith Model

IRF

By *implicit function theorem*, given an exo. shock $\mathrm{dZ} = (\mathrm{d}Z_0, \mathrm{d}Z_1 \ldots)'$ at $t = 0$, the linear impulse response of capital to this shock is given by:

$$d\mathrm{K} = -H_K^{-1} H_Z \, d\mathrm{Z}$$

where H_K and H_Z denote the Jacobians of H with respect to K and Z, evaluated at the steady state.

- Given dK, the impulse responses of other variables—for example, $\{Y_t, r_t, w_t\}_{t=0}^{\infty}$—follow immediately.
- In practice, $H_t(\mathrm{K}, \mathrm{Z}) = 0$ is solved up to a sufficiently large horizon T such that K and Z have approximately returned to their steady states by time T.

Example: The Krusell-Smith Model

Constructing the Jacobian

- We use the *chain rule* to relate the Jacobians H_K and H_Z to the derivatives of the $\mathcal{K}$.
- For example, differentiating $H_t(\mathrm{K}, \mathrm{Z}) = 0$ with respect to K_s
- The (t, s) entry of H_K is:

$$[H_K]_{t,s} = \frac{\partial \mathcal{K}_t}{\partial r_{s+1}} \frac{\partial r_{s+1}}{\partial K_s} + \frac{\partial \mathcal{K}_t}{\partial w_{s+1}} \frac{\partial w_{s+1}}{\partial K_s} - 1_{\{s=t\}}$$

- A similar expression applies to H_Z.
- In addition, the derivatives such as $\frac{\partial r_{s+1}}{\partial K_s}$ at (K^{ss}, Z^{ss}) can all be computed analytically.
- For example:

$$\frac{\partial r_{s+1}}{\partial K_s} = \alpha(\alpha - 1) Z^{ss} (K^{ss})^{\alpha-2} (N^{ss})^{1-\alpha}$$

Example: The Krusell-Smith Model

Constructing the Jacobian

Recall that:

$$d\mathrm{K} = -H_K^{-1} H_Z \, d\mathrm{Z},$$

To get $H_K^{-1} H_Z$, the only thing left is to solve the Jacobians of the $\mathcal{K}$ function with respect to its two inputs r and w.

$$[H_K]_{t,s} = \frac{\partial \mathcal{K}_t}{\partial r_{s+1}} \frac{\partial r_{s+1}}{\partial K_s} + \frac{\partial \mathcal{K}_t}{\partial w_{s+1}} \frac{\partial w_{s+1}}{\partial K_s} - 1_{\{s=t\}}$$

Fake News Algorithm

General Model Representation

SSJ development

SSJ in Continuous Time

To be developed

Determinacy and Large-Scale Solutions in the Sequence Space

To be developed

2nd SSJ

To be developed

fiscal and monetary policy

To be developed

behavioural macro

To be developed

life-cycle model

To be developed

dynamic spatial model

To be developed

production network

To be developed

het households + het firms

To be developed

het banks

To be developed

optimal policy

To be developed

macro labor

To be developed

Global Solution

Global Nonlinear Solutions in Sequence Space and the Generalized Transition Function

To be developed

DL in sequence space

To be developed

RL in sequence space

To be developed

Reference

Reference

To be developed